PRANAW GAUTAM

AI/ML Engineer | Full-stack Developer | Python, TypeScript & Flutter

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SUMMARY

AI/ML Engineer at Vibe6 Digital. Built the recommendation system for a client's audio-storytelling app, from raw event data and offline evaluation to an on-device Flutter prototype, and now ships full-stack features on a multi-tenant school-management SaaS (NestJS, Next.js, PostgreSQL with row-level security): online payments, assessment and data tooling. Proves results on held-out data, and proves tests can fail before trusting them. MCA, GPA 8.65/10. Open to full-time AI/ML and full-stack roles.

TECHNICAL SKILLS

Machine Learning: scikit-learn, recommender systems, collaborative filtering, content-based filtering, feature engineering, model evaluation (NDCG, precision/recall)

Data: Python, pandas, NumPy, SQL, EDA, statistical analysis, ETL pipelines, SQLAlchemy, Matplotlib, Seaborn

Apps & APIs: TypeScript, NestJS, Next.js, React, FastAPI, REST APIs, zod, Flutter, Dart

BI & Databases: Power BI (DAX), PostgreSQL (row-level security), Drizzle ORM, MySQL, Redis

Tools & Cloud: Git, GitHub Actions, pytest, Vitest, Playwright, Docker, BullMQ, AWS (EC2, S3)

EXPERIENCE

AI/ML Engineer | Vibe6 Digital LLP, Indore Aug 2026 – Present

Multi-tenant school-management SaaS (NestJS/Fastify, Next.js, PostgreSQL with forced row-level security)

- Top contributor to main in the first four weeks (251 of 582 commits): fees and payments, an integrations hub, transport tracking, attendance, exam marking and role dashboards.
- Shipped online fee payments through four gateways (Razorpay, Cashfree, PayU, Stripe), each school on its own account, with part payments and platform-admin control of which gateways schools may use; proved Cashfree and PayU end to end on their real sandboxes.
- Built assessment from exam defaults to results: papers for chosen students under one candidate rule reused by admit cards, marks, online tests, OMR and results; re-evaluation requests with a moderators' queue and notifications; teacher remarks; per-question difficulty from real results; marking schemes and question-bank tags.
- Added an optional, consent-first webcam identity check for online tests: direct uploads to private object storage through size-signed links, server-side JPEG verification, access limited to exam managers and automatic deletion after a retention period.
- Scaled student import from 2,000 to 20,000 rows with a staged, chunked, all-or-nothing import; wrote a dependency-free .xlsx export writer; secured the OMR recogniser with a shared secret.
- Built an integrations hub where each school connects its own payment gateway, GPS provider, APIs and signed webhooks, with encrypted credentials and outbound calls guarded against internal addresses.
- Tested against real PostgreSQL and object storage, broke each guard on purpose to prove its test fails (mutation testing), and added cross-tenant isolation checks for every new table.

Recommendation system for a client's audio-storytelling app (Python, scikit-learn, Flutter)

- Built the recommendation pipeline in Python (pandas, scikit-learn): cleaned a raw Firestore export, engineered user-title features and set up leakage-free, time-based splits.
- Benchmarked 8 approaches (popularity, TF-IDF content-based, item-item collaborative filtering, hybrid); weighted popularity beat heavier models on sparse data (NDCG@10 0.456 vs 0.419), steering the design to simpler ranking.
- Designed a shelf-based home screen that reached 87.4% of held-out users with a title they went on to play, vs 82.6% for a single list of equal length (McNemar p = 0.0018).
- Trained a logistic-regression ranker that lifted top-10 hit rate from 77.9% to 80.1% (562 users, p = 0.02).
- Caught an 18+ title being recommended to 99% of users who had not confirmed their age; enforced age gating in the ranking and shelf logic, cutting that to zero.

- Built the Flutter prototype that ranks titles on-device from exported model weights and explains every card; parity tests keep the Dart engine identical to the Python reference.

Data Science Intern | Shanti Infosoft LLP Jan 2026 – Jun 2026

- Built reusable preprocessing pipelines in Python (pandas, NumPy) as modular cleaning and transformation components.
- Ran exploratory data analysis to surface patterns and data-quality issues; presented findings with Matplotlib and Seaborn.
- Debugged and optimized pipeline scripts for reliability; took part in code reviews, updates and testing.

PROJECTS

Personalized Audio Recommender | Python, scikit-learn, FastAPI, pytest 2026 · Open source

- Open-source recommender on synthetic data: 7 models behind one interface, an explainable shelf home screen, a FastAPI service and demo, 79 tests with CI; tuned hybrid NDCG@10 0.177 vs 0.160 for popularity.

Vayu — Air Quality Forecasting | Python, DuckDB, LightGBM, FastAPI, React 2026 · Open source

- Normalised 71.9M CPCB station readings from 546 files and 113 header layouts into partitioned Parquet queried with DuckDB; implemented the official CPCB AQI and quarantined 22 stations whose weather sensors disagreed with reanalysis.
- 6-hour PM2.5 forecast with 35% lower error than persistence (MAE 41.4 vs 63.7 μg/m³) across 3 rolling-origin folds, detecting 66.5% of pollution-episode onsets.

Space Launch Analytics | Python, MySQL, SQLAlchemy, Power BI 2026

- Python ETL that validates and loads launch, rocket and ISRO data into a 4-table MySQL warehouse with engineered KPIs such as cost per kg and payload utilization.
- 6-page Power BI report with DAX on 7,500 launches since 1957: 83.05% success rate, 35 launch sites, 26 operators, cost trends and an ISRO deep-dive.

EDUCATION

Master of Computer Applications (MCA) | AKS University, Satna | GPA 8.65/10 2024 – 2026

Bachelor of Computer Applications (BCA) | APS University, Rewa 2015 – 2018

CERTIFICATIONS & ACHIEVEMENTS

- National Finalist (Top 10), GOER 2025 – Institute of Risk Management (India Affiliate) | RMAT score: 96% (240/250), 2025
- Advanced Program on Amazon Web Services (332 hours, Grade A1) – Anudip Foundation, supported by Accenture, 2025
- Data Science Bootcamp – GeeksforGeeks | Data Science Internship Certificate – Shanti Infosoft LLP, 2026